Scaled Relative Graph Toolbox

MATLAB Reference Documentation

Frequency-domain analysis of SISO and MIMO LTI systems
using Scaled Relative Graphs, Beltrami-Klein geometry,
and hyperbolic convex-hull methods.

Eder Baron-Prada, Adolfo Anta, Thomas Chaffey, Alberto Padoan.

Version 1.0

0 Contents

1 Introduction

The **Scaled Relative Graph (SRG) Toolbox** is a MATLAB toolbox for computing and visualizing the Scaled Relative Graph [6, 10] of linear time-invariant (LTI) systems. It supports both SISO and MIMO systems and provides tools for:

- Frequency-wise SRG computation in the Beltrami-Klein disk and the complex plane.
- Soft SRG via the hyperbolic convex hull of frequency-wise slices.
- Hard SRG for systems with unstable poles or non-minimum-phase zeros.
- Stability margin analysis and SRG-based feedback stability assessment.
- Frequency-wise circular (Q, S, R) multiplier extraction (solver-free).
- 2D and 3D visualization with a unified, theme-aware style system.
- Publication-ready figure export (PDF/EPS/PNG/TikZ).

1.1 Toolbox Structure

Lower-level helpers and the plotting/style infrastructure live in `bg/`, and the runnable example scripts live in `examples/`. Run `setup.m` once to add all three directories (root, `bg/`, `examples/`) to the MATLAB path.

```
tb/
|-- Contents.m                      % Function index (help Contents)
|-- setup.m                         % Add root + bg/ + examples/ to the path
|-- srg_compute.m                   % Core frequency-wise SRG computation
|-- srg_hard.m                      % Hard SRG boundary
|-- srg_homotopy.m                  % Homotopy of tau*G
|-- srg_scaled_graph.m              % Soft SRG (hyperbolic convex hull)
|-- srg_bounding_circle.m           % Minimum enclosing circle of an SRG set
|-- srg_qsr_multiplier.m            % Frequency-wise (Q,S,R) multiplier fit
|-- srg_export.m                    % Publication export (PDF/EPS/PNG/TikZ)
|-- srg_plot_gauss.m                % 2D Gauss-plane plot
|-- srg_plot_beltrami.m             % 2D Beltrami-Klein plot
|-- srg_plot_compare.m              % Multi-SRG overlay
|-- srg_plot_3d.m                   % 3D wire/filled plot
|-- srg_plot_3d_surface.m           % 3D surface plot
|-- srg_plot_3d_compare.m           % 3D comparison with intersection
|-- srg_plot_3d_beltrami.m          % 3D BK-disk plot
|-- srg_plot_3d_beltrami_compare.m  % 3D BK-disk comparison
|-- srg_plot_witness_3d.m           % Min-distance witness segments (3D)
|-- srg_stability_margin.m          % Frequency-wise stability margin
|-- examples/                       % Runnable demonstration scripts
|   |-- example_01_freqwise_srg.m   % Example: frequency-wise SRG
|   |-- example_02_soft_srg.m       % Example: soft SRG
|   |-- example_03_hard_srg.m       % Example: hard SRG
|   `-- example_04_qsr_export.m     % Example: QSR multiplier + export
`-- bg/                             % Internal helpers + plot/style backend
    |-- beltrami_inv.m
    |-- beltrami_map_matrix.m
    |-- beltrami_map_scalar.m
    |-- field_of_values.m
    |-- srg_refine_chords.m
    |-- srg_to_polyshape.m
    |-- srg_style.m
```

```
‘-- srg_apply_style.m
```

1.2 Quick Start

```
[caption={Minimal example: SISO SRG}]
G = tf([1 2], [1 3 2]);
[W, A, gplus, gminus, rawdata] = srg_compute(G, -3, 3, 200, 1);

figure;
srg_plot_gauss(G, gplus, gminus, 1, 200);
title('SRG of G');
```

```
[caption={Minimal example: MIMO feedback pair}]
G1 = [tf([1 2],[1 3 2])  tf([0.1],[1 1]);
      tf([0.05],[1 2])    tf([1],[1 1])];
G2 = [tf([1 1],[1 5])    tf([1],[3 2 2]);
      tf([1],[1 2])      -tf([0.5],[1 2])];

[~, ~, gp1, ~, rd1] = srg_compute(G1, -3, 3, 100, 500);
[~, ~, gp2, ~, rd2] = srg_compute(-inv(G2), -3, 3, 100, 500);

figure;
srg_plot_3d_compare(rd1, gp1, rd2, gp2, 0, 0, ...
    'Color1', 1, 'Color2', 5, 'FaceAlpha', 0.6);
```

2 Function Index

Function	Purpose
`srg_compute`	Core frequency-wise SRG computation
`srg_scaled_graph`	Soft SRG (hyperbolic convex hull)
`srg_hard`	Hard SRG for unstable/NMP systems
`srg_homotopy`	SRG of τG for multiple τ
`srg_stability_margin`	Frequency-resolved stability margin + witness pair
`srg_bounding_circle`	Minimum enclosing circle of an SRG set
`srg_qsr_multiplier`	Frequency-wise (Q, S, R) multiplier fit
`srg_export`	Publication export (PDF/EPS/PNG/TikZ)
`srg_plot_gauss`	2D SRG in Gauss plane
`srg_plot_beltrami`	2D SRG in Beltrami-Klein disk
`srg_plot_compare`	Multi-SRG overlay
`srg_plot_3d`	3D wire/filled plot
`srg_plot_3d_surface`	3D smooth surface
`srg_plot_3d_compare`	3D two-SRG comparison
`srg_plot_3d_beltrami`	3D BK-disk plot
`srg_plot_3d_beltrami_compare`	3D BK-disk comparison
`srg_plot_witness_3d`	3D min-distance witness segments
`srg_style`	Style configuration struct
`srg_apply_style`	Apply style to current axes

3 Mathematical Background

3.1 Gain, Phase, and the Scaled Relative Graph

For signals $u, y \in L_2^n$ with $u \neq 0$, define the **gain** and **phase** of the input-output pair as

$$\rho(u,y) := \frac{\|y\|}{\|u\|}, \qquad \theta(u,y) := \arccos\left(\frac{\langle u, y\rangle}{\|u\|\,\|y\|}\right), \tag{1}$$

with $\theta(u,y) = 0$ when $y = 0$. Each pair (ρ, θ) encodes a point $\rho\, e^{\pm j\theta} \in \mathbb{C}$: the modulus captures amplitude change and the argument captures the angular misalignment between input and output.

For a linear operator H, the **Scaled Relative Graph** (SRG) is the set of all such complex numbers over all non-zero inputs [9, 10]:

$$\mathrm{SRG}(H) := \left\{ \frac{\|Hu\|}{\|u\|}\, e^{\pm j\,\theta(u,Hu)} \;\middle|\; u \in L_2^n,\ u \neq 0 \right\}. \tag{2}$$

The SRG is always symmetric with respect to the real axis. Its modulus encodes gain variation across input directions and its argument encodes phase variation; together they generalize the Nyquist diagram to MIMO systems [6].

Frequency-wise decomposition for LTI systems

The Fourier transform diagonalizes convolution, so the input-output map of an LTI system $H(s) \in \mathcal{RH}_\infty^{n\times n}$ decouples across frequencies. The SRG at a single frequency $\omega \in \mathbb{R}$ is:

$$\mathrm{SRG}(H(j\omega)) := \left\{ \frac{\|H(j\omega)\,u\|}{\|u\|}\, e^{\pm j\,\arccos\left(\frac{\mathrm{Re}\{\langle H(j\omega)u,\,u\rangle\}}{\|H(j\omega)u\|\,\|u\|}\right)} \;\middle|\; u \in \mathbb{C}^n,\ u \neq 0 \right\} \subset \mathbb{C}. \tag{3}$$

The geometric complexity of (3) grows with system dimension n: for SISO systems ($n = 1$), each frequency slice is a conjugate pair of points; for $n = 2$, a pair of conjugate curves; for $n \geq 3$, a pair of conjugate filled regions.

The full operator SRG is the hyperbolic convex hull of all frequency-wise slices:

$$\mathrm{SRG}(H) = \mathrm{co}_h\Big(\bigcup_{\omega\in\mathbb{R}} \mathrm{SRG}(H(j\omega))\Big),$$

where co_h denotes the hyperbolic convex hull in the Beltrami-Klein model.

3.2 Beltrami-Klein (BK) Mapping

The toolbox maps the frequency-response matrix $G(j\omega)$ into the unit disk $\mathbb{D} = \{z \in \mathbb{C} : |z| \leq 1\}$ via the **Beltrami-Klein mapping**. For a matrix $T \in \mathbb{C}^{n\times n}$:

$$\Phi(T) = (I + T^*T)^{-1/2}(T^* - iI)(T - iI)(I + T^*T)^{-1/2}, \tag{4}$$

and for a scalar $\lambda \in \mathbb{C}$ (e.g., an eigenvalue):

$$f(\lambda) = \frac{(\bar\lambda - i)(\lambda - i)}{1 + \bar\lambda\lambda}. \tag{5}$$

The key geometric property of the BK model is that **Euclidean convex hulls in $\mathbb{D}$ coincide with hyperbolic convex hulls**, because hyperbolic geodesics in the BK model are Euclidean straight-line segments. This means that the soft SRG can be computed simply as the Euclidean convex hull of all frequency-wise BK-disk images.

3.3 Inverse BK Mapping

The inverse map g^{-1} recovers SRG points in the Gauss plane from BK disk coordinates via two branches:

$$g^{\pm}(z) = \frac{\mathrm{Im}(z) \pm i\sqrt{1-|z|^2}}{\mathrm{Re}(z)-1}, \tag{6}$$

where g^{+} gives the upper half-plane boundary and g^{-} the lower (conjugate) boundary of the SRG. The toolbox calls this map via `beltrami_inv`.

Because $g^{\pm}$ is strongly nonlinear, a straight chord in the BK plane does not, in general, map to a straight chord (or even a gently curved segment) in the Gauss plane. `field_of_values` returns only polygon vertices for normal matrices (the field of values of a normal matrix is exactly the convex hull of its eigenvalues), so BK-plane chords between vertices can be long. `srg_refine_chords` subdivides such chords, within the (linear) BK disk, before `beltrami_inv` is applied.

3.4 Soft SRG, Hard SRG, and Their Relationship

Three SRG variants appear in the toolbox, reflecting different classes of input signals and operator assumptions.

Frequency-wise SRG. The set (3) at a fixed frequency ω. Computed by `srg_compute` for each point in the frequency grid; stored in the `gplus`/`gminus` cell arrays.

Soft SRG [9]. Defined over L_2 signals and the full time horizon $[0,\infty)$; requires $u \in L_2^n$ and $y = Hu \in L_2^n$. For an LTI system in $\mathcal{RH}_\infty$, the soft SRG equals the hyperbolic convex hull of all frequency-wise slices:

$$\mathrm{SG}(H) := \mathrm{co}_h\left(\bigcup_{\omega\in\mathbb{R}} \mathrm{SRG}(H(j\omega))\right).$$

Computed by `srg_scaled_graph`.

Hard SRG [8]. Defined over extended L_{2e} signals using truncated gain and phase quantities ρ_T, θ_T over finite horizons $T > 0$. The hard SRG always satisfies $\mathrm{SG}(H) \subseteq \overline{\mathrm{SG}_e(H)}$. It accounts for:

- **Unstable poles**: max gain $\to \infty$, capped at a finite value `InfVal`.
- **Non-minimum-phase zeros**: min gain $= 0$.

Computed by `srg_hard`.

3.5 Feedback Stability Criterion

Consider the negative feedback interconnection of H_1 and H_2 shown in Figure 1.

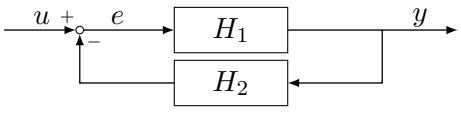

Figure 1: Negative feedback interconnection of H_1 and H_2.

The SRG stability theorem certifies L_2-stability via geometric separation of SRGs [2, 5]. Extensions to decentralized loops [1] and to power-electronics-dominated grids [3, 4] follow the same separation principle.

Frequency-wise GFT for LTI systems

For $H_1(s), H_2(s) \in \mathcal{RH}_\infty^{n\times n}$, the closed loop is L_2-stable if, at every frequency $\omega \in [0,\infty)$ and for all $\tau \in (0,1]$:

$$\mathrm{SRG}(H_1(j\omega))^{-1} \cap (-\tau\ \mathrm{SRG}(H_2(j\omega))) = \emptyset, \tag{7}$$

where $\mathrm{SRG}(H)^{-1} := \{z^{-1} \mid z \in \mathrm{SRG}(H),\ z \neq 0\}$. The homotopy parameter τ continuously deforms one SRG toward the origin; strict separation over the full range $\tau \in (0,1]$ certifies stability without requiring the full Nyquist encirclement count.

Toolbox convention

In practice, `srg_compute` is called on H_1^{-1} and $-H_2$ separately. The SRGs must be disjoint for the feedback pair to be stable. The stability margin (minimum distance between boundaries at each frequency) is computed by `srg_stability_margin`, and the closest-pair endpoints it returns can be drawn as witness segments by `srg_plot_witness_3d`.

Hard SRG stability

For operators that need not be L_2-stable, the hard separation condition

$$\mathrm{dist}(\mathrm{SG}_e^{-1}(H_1),\ -\mathrm{SG}_e(H_2)) > 0$$

is sufficient [8] and is evaluated geometrically from the `srg_hard` output via `fill` plots or `srg_plot_compare`.

4 Core Computation Functions

4.1 `srg_compute`

Purpose

Computes the frequency-wise SRG of an LTI system or constant matrix. This is the primary entry point for all SRG analyses.

Signature

```
[W, A, gplus, gminus, rawdata] = srg_compute(TransferFcn)
[W, A, gplus, gminus, rawdata] = srg_compute(TransferFcn, ...
    rangemin, rangemax, estpoints, points)
[...] = srg_compute(..., 'FreqScale', 'log'|'linear'|'auto')
[...] = srg_compute(..., 'Inverse', true|false)
[...] = srg_compute(..., 'Refine', true|false, 'RefineTol', 0.05)
```

Inputs

Argument	Type	Description
`TransferFcn`	`tf/ss/double`	Transfer function, state-space model, or constant matrix.
`rangemin`	`double`	Log base-10 of the minimum frequency [Hz] (or minimum linear frequency if `FreqScale='linear'`). Optional; defaults to automatic selection.

Argument	Type	Description
`rangemax`	`double`	Log base-10 of the maximum frequency [Hz] (or maximum linear frequency). Optional; defaults to automatic selection.
`estpoints`	`int`	Number of frequency evaluation points (default: `200`). Ignored when `FreqScale='auto'`, since the frequency vector is then determined by the system dynamics.
`points`	`int`	Angular resolution for the field of values (default: `64`). Set `1` for SISO (each slice is a single point). Use 32–2000 for MIMO depending on required smoothness. Also used as the chord-refinement density when `Refine=true` (see below).

Name-Value Arguments

Option	Default	Description
`FreqScale`	`'auto'`	Frequency spacing: `'log'` (logspace between `rangemin` and `rangemax`), `'linear'` (linspace), or `'auto'`. In `'auto'` mode the frequency vector is obtained from MATLAB's `nyquist` automatic selector.
`Inverse`	`false`	If `true`, returns the SRG of the *inverse* operator, $\mathrm{SRG}(T^{-1})$.
`Refine`	`true`	If `true`, subdivides long Beltrami-Klein-plane chords via `srg_refine_chords` before every call to `beltrami_inv`. Set to `false` to skip this step, e.g. for speed when the operator is known to be non-normal at every frequency in the sweep.
`RefineTol`	`0.05`	Maximum real- or imaginary-part step allowed between consecutive BK-plane boundary points before `srg_refine_chords` subdivides that chord.

Outputs

Output	Type	Description
`W`	`cell`	$1 \times N$ cell array; `W{i}` contains the BK-disk numerical range boundary at frequency i, after chord refinement (if `Refine=true`).
`A`	`cell`	$1 \times N$ cell array of eigenvalues in the BK disk.
`gplus`	`cell`	$1 \times N$ cell array; upper SRG boundary in the Gauss plane.
`gminus`	`cell`	$1 \times N$ cell array; lower SRG boundary in the Gauss plane.

Output	Type	Description
`rawdata`	`table`	Table with columns `freq`, `maxphase`, `minphase`, `norminf`, `norminfm`. When `Inverse=true`, an additional logical column `zero_interior` is included, flagging the frequency indices at which the computed inverse boundary contains non-finite points (i.e. where $\mathrm{SRG}(T^{-1})$ is unbounded; see the note below).

Constant matrices. When `TransferFcn` is a `double` matrix, the SRG is computed once and replicated across all frequency points. `gplus{i}` is identical for all i; the frequency axis in `rawdata` uses `logspace(-2,3,200)` by default and carries no physical significance.

Unbounded inverse SRGs. When `Inverse=true`, `srg_compute` checks at every frequency slice whether the resulting boundary (`gplus`/`gminus`) contains any non-finite points. In that case $\mathrm{SRG}(T^{-1})$ is (partially) unbounded and contains the point at infinity. A `warning('srg_compute:UnboundedInverseSRG', ...)` is then issued, reporting the frequency range(s) over which this occurs; the same information is recorded in `rawdata.zero_interior`.

Example

```
G = tf([1], [1 2 1]);
% Automatic frequency range
[W, A, gp, gm, rd] = srg_compute(G);
% Manual range
[W, A, gp, gm, rd] = srg_compute(G, -2, 3, 300, 1);
figure;
srg_plot_gauss(G, gp, gm, 1, length(rd.freq));
srg_plot_beltrami(G, W, A, 2, length(rd.freq));

% Inverse SRG, SRG(G^-1)
[Wi, A, gpi, gmi, rdi] = srg_compute(G, -2, 3, 300, 1, 'Inverse',
    true);
if any(rdi.zero_interior)
    disp('Inverse SRG is unbounded (contains infinity) at some
        frequencies.')
end

H = diag([3, -2/3, 1]);
[W, A, gp, gm, rd] = srg_compute(H, [], [], 1, 32, 'RefineTol', 0.02)
    ;
```

4.2 `srg_scaled_graph`

Purpose

Computes the **soft SRG** [9] (also called the Soft Scaled Graph) as the hyperbolic convex hull of all frequency-wise BK-disk slices.

Signature

```
[sg_plus, sg_minus, hull_bk] = srg_scaled_graph(W)
[sg_plus, sg_minus, hull_bk] = srg_scaled_graph([], ...
    'InputDomain', 'Gauss', 'GPlus', gplus, 'GMinus', gminus)
```

Inputs

Argument	Type	Description
`W`	`cell`	Cell array of BK-disk boundary curves from `srg_compute`. Pass `[]` when using Gauss-plane input.
`InputDomain`	`string`	`'BK'` (default) or `'Gauss'`.
`GPlus`	`cell`	Required when `InputDomain='Gauss'`.
`GMinus`	`cell`	Required when `InputDomain='Gauss'`.
`RefinementPoints`	`int`	Points per hull edge for the BK-to-Gauss mapping (default: 100).

Outputs

Output	Type	Description
`sg_plus`	`cell`	`{1}` cell containing the upper boundary of the soft SRG in the Gauss plane.
`sg_minus`	`cell`	`{1}` cell containing the lower boundary.
`hull_bk`	`cell`	`{1}` cell containing the complex vector of the convex hull boundary in the BK disk. Wrapped in a cell (rather than a bare vector) for direct compatibility with `srg_plot_beltrami` and `srg_plot_3d_beltrami`, which both dispatch on `numel(W)==1` to recognize a single static curve regardless of what is passed as `TransferFcn`.

Algorithm. (1) Pool all BK-disk points across frequencies. (2) Compute the Euclidean convex hull in the BK disk (= hyperbolic convex hull by the BK model property). (3) Interpolate hull edges (nonlinear inverse map requires dense sampling). (4) Apply g^{-1} to map to the Gauss plane.

4.3 `srg_hard`

Purpose

Computes the **hard SRG** [8] boundary of an LTI system, accounting for unstable poles and non-minimum-phase zeros.

Signature

```
[srg, srg_upper] = srg_hard(G, alphas, phis, frequencies)
[srg, srg_upper] = srg_hard(G, alphas, phis, frequencies, 'InfVal',
    50)
```

Inputs

Argument	Type	Description
`G`	`tf/ss`	Transfer function or state-space model.
`alphas`	`double`	Row vector of real base points for the alpha sweep. Typical: `linspace(-10, 10, 100)`.
`phis`	`double`	Row vector of angles in $[0, \pi]$ for boundary resolution. Typical: `linspace(0, pi, 200)`.
`frequencies`	`double`	Frequency grid in rad/s. Typical: `logspace(-3, 3, 500)`.
`InfVal`	`double`	Finite replacement for infinite gain (default: 10).

Outputs

Output	Type	Description
`srg`	`complex`	Full SRG boundary (upper + mirrored lower half).
`srg_upper`	`complex`	Upper half-plane boundary only.

Algorithm

1. For each α_i in `alphas`: construct $G(s) - \alpha_i I$ and compute $\sigma_{\min}$, $\sigma_{\max}$ via SVD sweep. If $G - \alpha_i I$ is unstable, set $\sigma_{\max} =$ `InfVal`. If $G - \alpha_i I$ has non-minimum-phase zeros, set $\sigma_{\min} = 0$.
2. Find the real interval $[a, b]$ contained in all max-gain circles $[\alpha_i - r_i^{\max}, \alpha_i + r_i^{\max}]$.
3. For each angle ϕ in `phis`: intersect all max-gain circles centered at α_i to find the tightest outer radius from $z_0 = (a + b)/2$.
4. Carve out min-gain circles from the resulting contour.
5. Mirror the upper boundary to obtain the full boundary.

Example

```
s = tf('s');
G = (s+7)/(s-1);     % unstable pole

alphas = linspace(-10, 10, 100);
phis   = linspace(0, pi, 200);
freqs  = logspace(-3, 3, 500);

[srg_bnd, ~] = srg_hard(G, alphas, phis, freqs, 'InfVal', 100);

figure;
srg_plot_compare(srg_bnd, 'Names', "Hard SRG", 'Mode', 'filled');
```

4.4 `srg_homotopy`

Purpose

Computes frequency-wise SRGs of the scaled system $G_\tau = \tau \cdot G$ for multiple τ values. Useful for visualizing how the SRG shrinks toward the origin as $\tau \to 0$.

Signature

```
homotopy_data = srg_homotopy(G, tau_values, rangemin, rangemax, ...
                             estpoints, points)
```

Output

A struct array with one element per τ value. Each element has fields: `tau`, `W`, `A`, `gplus`, `gminus`, `rawdata`.

Example

```
G   = tf([1 2], [1 3 2]);
hd = srg_homotopy(G, [1.0, 0.6, 0.3], -3, 3, 200, 64);

figure;
for k = 1:numel(hd)
    srg_plot_3d(hd(k).rawdata, hd(k).gplus, 0, 0, k, 'FaceAlpha',
       0.6);
    hold on;
end
```

4.5 `srg_stability_margin`

Purpose

Computes the frequency-resolved stability margin: the minimum Euclidean distance between two SRG boundaries at each frequency, together with the closest-pair endpoints on each boundary (the witness pair). Pairwise distances are computed without the Statistics Toolbox via a `bsxfun`-based expansion.

Signature

```
[stab_margin, freq, x1, x2] = srg_stability_margin(gplus1, gplus2,
   rawdata)
```

Inputs / Outputs

Name	Type	Description
`gplus1/2`	`cell`	SRG boundaries from `srg_compute`. Accepts scalar cells (constant operator) or frequency-indexed cells.
`rawdata`	`table/vector`	Frequency vector or table from `srg_compute`.
`stab_margin`	`double`	Minimum distance per frequency.
`freq`	`double`	Frequency vector.
`x1, x2`	`complex`	Closest points on each SRG at every frequency (witness pair).

4.6 `srg_bounding_circle`

Purpose

Computes the minimum enclosing circle of a set of complex SRG boundary points. Because SRGs are symmetric about the real axis, the center is constrained to the real line; the problem

reduces to a 1D minimax solved with `fminbnd` (base MATLAB, no toolboxes).

Signature

```
[center, radius] = srg_bounding_circle(srg)
[center, radius] = srg_bounding_circle(srg, 'Plot', true)
```

Name	Type	Description
`srg`	`complex`	Column vector of SRG boundary points.
`Plot`	`logical`	Overlay the circle on current axes (default: false).
`center`	`double`	Real-valued circle center.
`radius`	`double`	Minimum enclosing radius.

4.7 `srg_qsr_multiplier`

Purpose

Extracts a frequency-wise circular (Q, S, R) multiplier from the SRG. At each frequency the tightest circular region $\Pi(c, r)$ enclosing the SRG slice is found via `srg_bounding_circle`; rational transfer functions are then fit to the parameter sequences $c(\omega)$ and $R(\omega) = r^2 - c^2$. Fitting uses Sanathanan-Koerner (SK) weighted least squares with AIC order selection.

For the interior disk $\mathbb{D}^{\text{int}}(c, r)$, the circular multiplier (Proposition 1 of [5]) is

$$\Pi_{\text{int}}(c, r) = \begin{bmatrix} -1 & c \\ c & r^2 - c^2 \end{bmatrix}, \qquad Q = -1,\ S = c,\ R = r^2 - c^2, \tag{8}$$

and Π_{int} is positive-negative (soft-hard equivalent) iff $r > |c|$.

Signature

```
out = srg_qsr_multiplier(gplus, rawdata)
out = srg_qsr_multiplier(gplus, rawdata, 'MaxOrder', 8, ...
                         'SKIter', 20, 'Plot', true)
```

Name-Value Arguments

Option	Default	Description
`MaxOrder`	8	Maximum TF order searched (AIC selects within 1…MaxOrder).
`SKIter`	20	SK iterations per candidate order.
`Plot`	`false`	Diagnostic plot of c, R data + fits, and $r > \|c\|$.

Output struct fields

Field	Type	Description
`freq`	`double`	Frequency vector [Hz].
`c`	`double`	Per-frequency circle center.
`r`	`double`	Per-frequency circle radius.

Field	Type	Description
`R`	`double`	Per-frequency $R = r^2 - c^2$.
`pos_neg`	`logical`	True where $r > \|c\|$ (multiplier is positive-negative).
`S_tf`	`tf`	Fitted TF for $c(\omega)$ (S parameter).
`R_tf`	`tf`	Fitted TF for $R(\omega)$ (R parameter), signed-absolute overbounded.
`S_ord`, `R_ord`	`int`	Selected TF orders.
`R_scale`	`double`	Multiplicative overbound factor $k \geq 1$ applied to R_tf.
`Pi`	`struct`	Multiplier struct with `Q=-1`, `S=S_tf`, `R=R_tf`.

Example

```
G = tf([1], [1 2 1]);
[~,~,gp,~,rd] = srg_compute(G, -2, 3, 200, 32);
out = srg_qsr_multiplier(gp, rd, 'Plot', true);
fprintf('Selected orders: S=%d  R=%d\n', out.S_ord, out.R_ord);
```

5 Visualization Functions

All plotting functions share a common set of behaviors:

- **SISO detection** is data-driven: if every cell in `gplus` contains a single unique complex value, the system is treated as SISO regardless of the `TransferFcn` type.
- **Constant matrices** (`double` inputs to `srg_compute`) produce a static SRG plotted once.
- **Theming** is controlled via the `'Theme'` option (see Section 6).
- All functions call `hold on` internally so they can be layered in a single figure.

5.1 `srg_plot_gauss`

Plots the SRG in the Gauss (complex) plane.

```
srg_plot_gauss(TransferFcn, gplus, gminus, color, estpoints)
srg_plot_gauss(..., 'Fill', true, 'FaceAlpha', 0.5, 'Theme', 'dark')
```

Option	Default	Description
`color`	—	Color index (1–8) or RGB triplet.
`Fill`	`false`	Fill the SRG interior. For SISO: fills the region swept by the gplus/gminus traces. For MIMO: fills each frequency slice.
`FaceAlpha`	0.5	Fill transparency.
`Theme`	`'default'`	Style theme.

5.2 `srg_plot_beltrami`

Plots the SRG in the Beltrami-Klein disk.

```
srg_plot_beltrami(TransferFcn, W, A, color, estpoints)
srg_plot_beltrami(..., 'Fill', true, 'FaceAlpha', 0.4)
```

For SISO systems, draws the BK trajectory of the scalar across frequency as a continuous curve inside the disk with frequency markers.

Static curves (e.g. `srg_scaled_graph` output). A single already-computed boundary curve is detected and plotted as one continuous curve whenever `numel(W)==1` – regardless of what `TransferFcn` is passed for color/bookkeeping purposes. This mirrors `srg_plot_3d_beltrami`'s convention (`numel(gplus1)==1`), so callers no longer need to pass a placeholder `double` as `TransferFcn` just to get single-curve behavior – passing the actual system object works directly.

5.3 `srg_plot_compare`

Overlays multiple SRGs with automatic color cycling and a legend. Accepts either:

- **Cell arrays** (`gplus` from `srg_compute`): plots per-frequency curves or SISO trajectories.
- **Complex vectors** (`srg` from `srg_hard`): plots as filled `polyshape` regions.

```
srg_plot_compare(srg1, srg2, ...)
srg_plot_compare(srg1, srg2, 'Names', ["Plant","Controller"], ...
    'Fill', true, 'FaceAlpha', 0.6, 'Theme', 'publication')
```

Option	Default	Description
`Names`	auto	String array of legend entries.
`Theme`	`'default'`	Style theme.
`Mode`	`'filled'`	`'filled'` or `'boundary'` (for vector inputs).
`FaceAlpha`	from theme	Fill transparency.
`Fill`	`false`	Fill interior of cell-array inputs (MIMO slices or SISO region).
`Title`	`''`	Plot title.

No intersection detection. Each of the N SRGs is plotted independently – SISO/MIMO detection happens per input, so mixed SISO/MIMO overlays already work without special handling – but unlike the `_compare` 3D functions below, this function does not compute or highlight where two SRGs cross or overlap; it is a purely qualitative overlay.

5.4 `srg_plot_3d`

3D wire or filled plot of SRG boundaries stacked along the frequency axis (log-scaled z-axis).

```
srg_plot_3d(rawdata, gplus, fmin, fmax, color)
srg_plot_3d(..., 'Fill', true, 'FaceAlpha', 0.4, 'Mirror', true, ...
            'FillTightness', 0, 'ShowBoundary', true)
```

Option	Default	Description
`fmin/fmax`	0	Frequency window (0 = full range).
`Mirror`	`true`	Plot the conjugate (lower) half as well.
`Fill`	`false`	Fill each frequency slice (MIMO only; warning issued for SISO).
`FaceAlpha`	0.3	Patch transparency.
`ShowBoundary`	`true`	Overlay a wire boundary when `Fill=true`.

Option	Default	Description
`BoundaryColor`	same as fill	Wire color when filled.
`FillTightness`	0	0 = convex hull per slice; 1 = exact polyshape boundary.
`gminus`	`{}`	Lower boundary cell array (SISO only).

5.5 `srg_plot_3d_surface`

Renders the SRG as a continuous smooth `surf` across frequency. Each slice is resampled to a common number of arc-length-parameterized points; adjacent slices are rotationally aligned before stitching.

```
srg_plot_3d_surface(rawdata, gplus, fmin, fmax)
srg_plot_3d_surface(..., 'Color', 2, 'FaceAlpha', 0.6, ...
    'EdgeAlpha', 0.1, 'Mirror', true, 'NumPoints', 100, ...
    'Lighting', true, 'MinDiameter', -1)
```

Option	Default	Description
`Color`	1	Color index or RGB triplet.
`FaceAlpha`	0.6	Surface transparency.
`EdgeAlpha`	0.1	Edge transparency.
`Mirror`	`true`	Also render the conjugate surface.
`NumPoints`	100	Resampling points per slice.
`Lighting`	`true`	Enable Gouraud lighting.
`MinDiameter`	-1	Skip slices smaller than this (auto-detected when -1).

5.6 `srg_plot_3d_compare`

Overlays two SRG surfaces in 3D and highlights frequency slices where the two SRGs intersect or touch.

```
srg_plot_3d_compare(rawdata1, gplus1, rawdata2, gplus2, fmin, fmax)
srg_plot_3d_compare(..., 'Color1', 1, 'Color2', 5, ...
    'FaceAlpha', 0.45, 'EdgeAlpha', 0.5, 'Lighting', true, ...
    'IntersectColor', [], 'IntersectAlpha', 0.9, ...
    'Mirror', true, 'NumPoints', 100, 'MinDiameter', -1, ...
    'IntersectTol', -1)
```

Option	Default	Description
`IntersectTol`	-1 (auto)	Maximum boundary-to-boundary distance still counted as a crossing, for cases where the two SRGs touch or cross without enclosing a shared area (see below). Auto-scaled from the median magnitude of both boundaries.

5.7 `srg_plot_3d_beltrami`

3D wire plot in the Beltrami-Klein disk across frequencies.

```
srg_plot_3d_beltrami(rawdata, W, fmin, fmax, color)
srg_plot_3d_beltrami(..., 'Theme', 'dark')
srg_plot_3d_beltrami(..., 'ShowDisk', true, 'DiskAlpha', 0.06, ...
    'DiskColor', [], 'DiskNTheta', 180)
```

Handles three cases automatically: constant matrix (replicate single curve at every frequency level), SISO (trajectory of scalar BK image), MIMO (per-frequency curve stack).

Option	Default	Description
`ShowDisk`	`true`	Render the unit-circle as a translucent cylinder spanning the plotted frequency range.
`DiskAlpha`	0.06	Disk surface transparency.
`DiskColor`	gray	Disk color (RGB or index).
`DiskNTheta`	180	Angular resolution of the cylinder.

5.8 `srg_plot_3d_beltrami_compare`

Overlays two SRG surfaces in the BK disk in 3D, with an optional unit-disk cylinder boundary and intersection/crossing highlighting.

```
srg_plot_3d_beltrami_compare(rawdata1, W1, rawdata2, W2, fmin, fmax)
srg_plot_3d_beltrami_compare(..., ...
    'Color1', 4, 'Color2', 5, 'FaceAlpha', 0.8, ...
    'Lighting', true, 'IntersectColor', 1, 'IntersectAlpha', 1, ...
    'ShowDisk', true, 'DiskAlpha', 0.06, 'DiskColor', [], ...
    'DiskNTheta', 180, 'NumPoints', 100, 'MinDiameter', -1, ...
    'IntersectTol', -1)
```

Option	Default	Description
`ShowDisk`	`true`	Render the unit-circle as a translucent cylinder.
`DiskAlpha`	0.06	Disk surface transparency.
`DiskColor`	gray	Disk color (RGB or index).
`DiskNTheta`	180	Angular resolution of the cylinder.
`IntersectTol`	-1 (auto)	Maximum boundary-to-boundary distance still counted as a crossing when the two SRGs touch without enclosing a shared area. Auto-scaled from the median magnitude of both boundaries.

5.9 `srg_plot_witness_3d`

Overlays the minimum-distance witness segment(s) on an existing 3D SRG plot. Call after `srg_plot_3d_compare` on the same axes, using the closest-pair endpoints returned by `srg_stability_margin`.

```
srg_plot_witness_3d(x1, x2, freq, stab_margin)
srg_plot_witness_3d(..., 'Stride', 10, 'Mirror', true, ...
    'Color', 8, 'LineWidth', 2, 'MarkerSize', 10)
```

By default only the single critical segment (at the frequency of minimum margin) is drawn, with both endpoints marked. The `Stride` option additionally draws every `Stride`-th background segment in thin grey; rendering every segment would produce a misleading surface-like artifact.

Option	Default	Description
`Stride`	0	Draw every k-th background segment (0 = critical only).
`Mirror`	`true`	Also draw conjugate lower-half segments.
`Color`	8	Segment color (palette index or RGB).
`LineWidth`	2	Witness segment line width.
`MarkerSize`	10	Endpoint marker size.

6 Style System and Figure Export

6.1 `srg_style`

Returns a style struct `s` consumed by all plot functions.

```
s = srg_style()                          % default theme
s = srg_style('Theme', 'dark')
s = srg_style('Theme', 'publication')
```

Available Themes

Theme	Description
`default`	White background, Helvetica font, minor grid.
`light`	Identical to default (alias for clarity).
`dark`	Dark background (`[0.15 0.15 0.18]`), pale CB colors, white grid lines.
`publication`	White background, Times New Roman, 14 pt, LaTeX interpreter, `FaceAlpha=0.35`.

Struct Fields

Field	Type	Description
`s.palette`	$N \times 3$	CB colorblind-safe colors used for cycling (8×3 for `default/light/publication`; the `dark` theme uses a 7×3 set of pale variants).
`s.color(n)`	function	Returns `s.palette(mod(n-1,size(s.palette,1))+1,:)`.
`s.linewidth`	double	Default line width.
`s.fontsize`	double	Label font size.
`s.fontname`	string	Font name.
`s.facealpha`	double	Default fill transparency.
`s.interpreter`	string	`'tex'` or `'latex'`.
`s.grid`	string	`'minor'` or `'on'`.
`s.axiscolor`	RGB	Color for axis crosshairs.
`s.bgcolor`	RGB	Axes background color.

Field	Type	Description
`s.cb.*`	RGB	Direct access to all CB color variants (`s.cb.red`, `s.cb.pale_blue`, `s.cb.dark_green`, etc.).

Colorblind-Safe Palette (Primary Colors)

Index	Name	RGB
1	CB Blue	$[0.267, 0.467, 0.667]$
2	CB Red	$[0.933, 0.400, 0.467]$
3	CB Green	$[0.133, 0.533, 0.200]$
4	CB Yellow	$[0.800, 0.733, 0.267]$
5	CB Purple	$[0.667, 0.200, 0.467]$
6	CB Cyan	$[0.400, 0.800, 0.933]$
7	CB Grey	$[0.733, 0.733, 0.733]$
8	CB Orange	$[0.850, 0.325, 0.098]$

6.2 `srg_apply_style`

Applies the style struct to the current axes.

```
srg_apply_style(s)
srg_apply_style(s, 'Crosshairs', true, 'AxisEqual', true, ...
                   'Labels', {'Re', 'Im'})
```

6.3 `srg_export`

Purpose

Exports the current figure in a publication-ready format, re-applying the `publication` theme from `srg_style` before writing. The file extension selects the format.

Signature

```
srg_export(filename)
srg_export(filename, 'Width', 8.5, 'Height', 6.5, 'DPI', 300, ...
           'Fig', gcf, 'ApplyStyle', true, 'FontSize', [])
```

Option	Default	Description
`Fig`	`gcf`	Figure handle to export (call `figure(fig)` first).
`Width`	8.5 cm	Output width (8.5 cm = one IEEE column; 17.4 cm = double).
`Height`	6.5 cm	Output height.
`DPI`	300	Resolution for raster (.png) export.
`ApplyStyle`	`true`	Re-apply the publication style before export.
`FontSize`	`[]`	Override font size; `[]` uses the theme default.

Formats by extension

Ext	Backend
`.png`	`exportgraphics` at the specified DPI (crops to plot content), with a `getframe+imwrite` fallback if `exportgraphics` is unavailable (no print driver required).
`.tikz`	`matlab2tikz`; `\input{}`-ready, inherits the LaTeX document font. Requires `matlab2tikz` on the path.
`.pdf`	Best-effort vector PDF.
`.eps`	Best-effort vector EPS.

Example

```
figure(fig);     % make the target figure current first
[~,~,gp,gm,~] = srg_compute(G, -2, 3, 200, 1);
srg_plot_gauss(G, gp, gm, 1, 200, 'Theme', 'publication');
srg_export('fig_srg.png', 'DPI', 300);      % always works
srg_export('fig_srg.tikz');                 % vector via pdflatex
srg_export('fig_srg.pdf', 'Width', 8.5);  % best-effort vector PDF
```

7 Example Scripts

7.1 Example 1 — Frequency-wise SRG (`example_01_freqwise_srg.m`)

Demonstrates the basic `srg_compute` → `srg_plot_*` → `srg_stability_margin` pipeline for three feedback pairs of increasing complexity. The MIMO loop also shows the homotopy overlay and the witness segment via `srg_plot_witness_3d`.

Loop A — SISO. Two stable systems; `points=1`. All 2D and 3D plots are demonstrated.
Loop B — MIMO 2×2 (stable). `points=500`. Includes a homotopy overlay and a 3D witness segment.
Loop C — MIMO 2×2 (unstable closed loop). `points=500`. Demonstrates SRG-based detection of instability via non-empty intersection of the two SRGs.

7.2 Example 2 — Soft SRG (`example_02_soft_srg.m`)

Demonstrates `srg_scaled_graph` for both SISO and MIMO systems, including comparison of frequency-wise slices versus the hyperbolic convex hull.

7.3 Example 3 — Hard SRG (`example_03_hard_srg.m`)

Demonstrates `srg_hard` for:

Case A — MIMO 2×2. G_1 has an unstable pole at $s = +1$.
Case B — MIMO 3×3. G_2 has an unstable pole and an integrator.

7.4 Example 4 — QSR Multiplier and Export (`example_04_qsr_export.m`)

Demonstrates the multiplier and export layers built on top of the core computation.

Part 1 — QSR multiplier. Fits the circular (Q, S, R) multiplier of a stable MIMO 2×2 plant with `srg_qsr_multiplier` (`Plot` on, `MaxOrder = 10`, `SKIter = 20`), reporting the selected S/R orders, the overbound factor k, and how often $r > |c|$ holds across frequencies.
Part 2 — Bounding circle. Visualizes the minimum enclosing circle of the lowest-frequency SRG slice via `srg_bounding_circle`, overlaid on the corresponding `srg_plot_gauss` slice.

Part 3 — 3-D surface and export. Renders the full 3-D SRG surface with `srg_plot_3d_surface`, overlays the per-frequency minimum enclosing circle at every slice, then exports the figure to PNG (reliable, no print driver required), TikZ (vector, requires `matlab2tikz`), and PDF (best-effort, requires a working MATLAB print driver) via `srg_export`.

8 Conventions and Data Format

8.1 Cell Array Convention

All computation functions return cell arrays indexed by frequency:

- **SISO:** `gplus{i}` is a vector of identical complex scalars (field of values of a 1×1 matrix). The representative value is `gplus{i}(1)`.
- **MIMO:** `gplus{i}` is a complex vector tracing the boundary of the numerical range at frequency i.
- **Constant matrix:** `gplus{i}` is identical for all i; the cell array has `estpoints` copies of the same curve.

8.2 Frequency Convention

`srg_compute` (and therefore `srg_homotopy` and the `freq` column consumed by `srg_qsr_multiplier`) uses frequencies in **Hz** (`freqresp` with the `'Hz'` unit). `srg_qsr_multiplier` converts internally to rad/s ($\omega = 2\pi f$) for the rational fit. `srg_hard` uses frequencies in **rad/s** (`freqresp` with `1i*omega`).

8.3 Feedback Partner Convention

For the feedback stability criterion, the two SRGs to overlay are:

- **Plant partner:** pass `G1` directly.
- **Controller partner:** pass `-inv(G2)`.

The SRGs must be disjoint for stability.

8.4 QSR Multiplier Format

`srg_qsr_multiplier` returns `out.Pi` as a struct with `Q = -1` (scalar), `S = out.S_tf` and `R = out.R_tf` (transfer functions). These correspond to the block multiplier (8) with the convention $\Pi = \left[\begin{smallmatrix} Q & S \\ S & R \end{smallmatrix}\right]$.

8.5 Output Table Fields (`rawdata`)

Field	Type	Description
`freq`	double	Frequency vector [Hz], column.
`maxphase`	double	Max singular angle of g^- [deg].
`minphase`	double	Min singular angle of g^- [deg].
`norminf`	double	$\|g^+\|_\infty$ (max gain).
`norminfm`	double	$\|g^+\|_{-\infty}$ (min gain).

9 Dependencies and Licenses

9.1 Third-Party Code

File	Author	License
`field_of_values.m`	N. J. Higham [7]	Matrix Computation Toolbox
`SrgTools.jl`	Julius P. J. Krebbekx [8]	BSD 3-Clause
`matlab2tikz` (optional)	N. Schlömer et al.	MIT (external, not bundled)

9.2 MATLAB Requirements

- MATLAB R2020b or later (required for `arguments` blocks).
- Control System Toolbox (`freqresp`, `isstable`, `tzero`, `tf`, `ss`, `feedback`).
- `matlab2tikz` is required only for the `.tikz` branch of `srg_export`.

9 References

[1] Eder Baron-Prada, Adolfo Anta, and Florian Dörfler. On decentralized stability conditions using scaled relative graphs. *IEEE Control Systems Letters*, 9:691–696, 2025.

[2] Eder Baron-Prada, Adolfo Anta, Alberto Padoan, and Florian Dörfler. Stability results for MIMO LTI systems via scaled relative graphs, 2025. URL https://arxiv.org/abs/2503.13583. To appear in IFAC World Congress 2026.

[3] Eder Baron-Prada, Adolfo Anta, and Florian Dörfler. On the impact of operating points on small-signal stability: Decentralized stability sets via scaled relative graphs, 2026. URL https://arxiv.org/abs/2603.13922. arXiv:2603.13922.

[4] Eder Baron-Prada, Adolfo Anta, and Florian Dörfler. Stability analysis of power-electronics-dominated grids using scaled relative graphs. *IEEE Transactions on Power Systems*, pages 1–15, 2026. doi: 10.1109/TPWRS.2026.3674752.

[5] Eder Baron-Prada, Julius P. J. Krebbekx, Adolfo Anta, and Florian Dörfler. Scaled graph containment for feedback stability: Soft–hard equivalence and conic regions, 2026. URL https://arxiv.org/abs/2604.05567. arXiv:2604.05567.

[6] Thomas Chaffey, Fulvio Forni, and Rodolphe Sepulchre. Scaled relative graphs for system analysis. In *Proceedings of the 60th IEEE Conference on Decision and Control (CDC)*, pages 3087–3094, 2021. doi: 10.1109/CDC45484.2021.9683133. URL https://arxiv.org/abs/2103.13971.

[7] Nicholas J. Higham. The Matrix Computation Toolbox. URL http://www.ma.man.ac.uk/~higham/mctoolbox. http://www.ma.man.ac.uk/~higham/mctoolbox.

[8] Julius P. J. Krebbekx, Eder Baron-Prada, Roland Tóth, and Amritam Das. Computing the hard scaled relative graph of LTI systems, 2025. URL https://arxiv.org/abs/2511.17297. arXiv:2511.17297.

[9] Richard Pates. The scaled relative graph of a linear operator. *arXiv preprint*, 2021. URL https://arxiv.org/abs/2106.05650. arXiv:2106.05650.

[10] Ernest K. Ryu, Robert Hannah, and Wotao Yin. Scaled relative graph: Nonexpansive operators via 2D Euclidean geometry, 2019. URL https://arxiv.org/abs/1902.09788. arXiv:1902.09788.